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Work done, $W = F\,dr\cos\theta$

$$W = 25 \times 15 \times \cos 30 = 25 \times 15 \times \frac{\sqrt{3}}{2}$$

$$W = 324.76 \text{ J}$$

4.1.2 Work done by a constant force

When a constant force F acts on a body, the small work done (dW) by the force in producing a small displacement dr is given by the relation,

$$dW = (F\cos\theta)\, dr \qquad (4.3)$$

The total work done in producing a displacement from initial position $\mathbf{r}_i$ to final position $\mathbf{r}_f$ is,

$$W = \int_{r_i}^{r_f} dW \qquad (4.4)$$

$$W = \int_{r_i}^{r_f} (F\cos\theta)\, dr = (F\cos\theta)\int_{r_i}^{r_f} dr =$$

$$(F\cos\theta)(r_f - r_i) \qquad (4.5)$$

The graphical representation of the work done by a constant force is shown in Figure 4.5. The area under the graph shows the work done by the constant force.

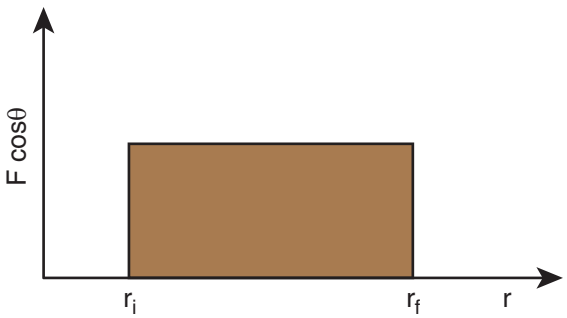
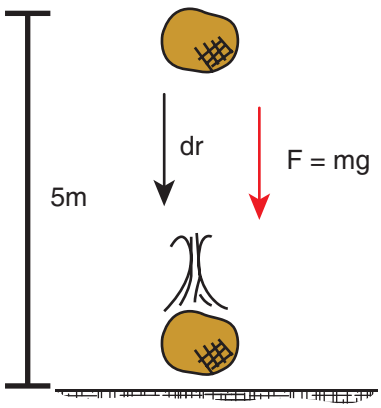


Figure 4.5 Work done by the constant force

EXAMPLE 4.2

An object of mass 2 kg falls from a height of 5 m to the ground. What is the work done by the gravitational force on the object? (Neglect air resistance; Take g = 10 m s^{-2})



Solution

In this case the force acting on the object is downward gravitational force $m\vec{g}$. This is a constant force.

Work done by gravitational force is

$$W = \int_{r_i}^{r_f} \vec{F}\cdot d\vec{r}$$

$$W = (F\cos\theta)\int_{r_i}^{r_f} dr = (\,mg\cos\theta)\,(r_f - r_i).$$

The object also moves downward which is in the direction of gravitational force ($\vec{F} = m\vec{g}$) as shown in figure. Hence, the angle between them is $\theta = 0°$; $\cos 0° = 1$ and the displacement, $(r_f - r_i) = 5\,\text{m}$

$$W = mg(r_f - r_i)$$

$$W = 2 \times 10 \times 5 = 100$$

The work done by the gravitational force on the object is positive.